

4052 MATH P2 SOLN.pdf

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Math
P2.pdf

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 MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER INDEX NUMBER

MATHEMATICS **4052/02**
Paper 2 October/November 2023
2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE ON ANY BARCODES.

Answer all questions.
The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.
Omission of essential working will result in loss of marks.
The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π , use either your calculator value or 3.142.

This document consists of 21 printed pages and 3 blank pages.

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Mathematical Formulae

Compound interest

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2}ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2}r^2\theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$



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TURN OVER FOR QUESTION 1





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- 1 (a) The table shows the resident population of Singapore in three years.

Year	2000	2010	2020
Resident population	3 273 363	3 771 721	4 044 210

- (i) Write the 2010 resident population in standard form, correct to three significant figures.

Answer 3.77×10^6 [1] - B1

- (ii) Calculate the percentage increase in the resident population from 2000 to 2020.

$$\frac{4044210 - 3273363}{3273363} \times 100\% = 23.5\% \text{ (3 s.f.)}$$

Answer 23.5% [2] - M1 - A1

- (iii) In 2000, 18.7% of the total population of Singapore was non-resident.

Calculate the total population of Singapore in 2000.

$$\text{Total population} = \frac{100}{100 - 18.7} \times 3273363$$

≈ 4026277 (nearest whole no.)
Answer 4026277 [2] - M1 - A1
(≈ 4026000 to 3 s.f.)

- (b) Population density is defined as the number of people per square kilometre.
In 2020, the population density of Singapore was 7810 people per square kilometre.
In 2020, the total population of Singapore was 5.69 million.

Calculate the area of Singapore.

$$\text{area} = \frac{5.69 \times 10^6}{7810} = 729 \text{ km}^2 \text{ (3 s.f.)}$$

Answer 729 km^2 [2] - M1 - A1



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- (c) The diameter of a red blood cell is 7.8×10^{-6} m.
In a scale drawing, the diameter of a red blood cell is 3.9 cm.

- (i) Find the scale used for the drawing.
Give your answer in the form $n : 1$.

$$3.9 \text{ cm} = 0.039 \text{ m} \quad \text{--- M1}$$

$$\begin{aligned} & \div 7.8 \times 10^{-6} \quad 0.039 : 7.8 \times 10^{-6} \\ & 5000 : 1 \quad \div 7.8 \times 10^{-6} \quad 5000 \quad \text{--- A1} \\ & \text{Answer } \dots\dots\dots : 1 \quad [2] \end{aligned}$$

- (ii) Drawn to the same scale as the red blood cell, the diameter of a blood platelet is 8 mm.

Find the actual diameter of the blood platelet.
Give your answer in millimetres in standard form.

Actual diameter

$$= \frac{8}{5000}$$

$$= 1.6 \times 10^{-3} \text{ mm}$$

$$\text{Answer } \dots\dots\dots 1.6 \times 10^{-3} \quad \text{--- B1} \quad \text{mm} \quad [1]$$



- 2 (a) Solve $6-7x = 5(1-2x)$.

$$6-7x = 5-10x \quad \text{--- M1: expand}$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

$$\text{Answer } x = -\frac{1}{3} \quad \text{--- A1} \quad [2]$$

- (b) Solve the inequality $5-y > 7$.

$$-y > 2$$

$$y < -2$$

$$\text{Answer } y < -2 \quad \text{--- B1} \quad [1]$$

(c) $c = \frac{1}{b} + \frac{a}{3-a}$

- (i) Find c when $a = 2$ and $b = 4$.

$$c = \frac{1}{4} + 2$$

$$= 2\frac{1}{4}$$

$$\text{Answer } c = 2\frac{1}{4} \quad [1]$$

- (ii) Rearrange the formula to make a the subject.

$$c = \frac{1}{b} + \frac{a}{3-a}$$

$$\frac{a}{3-a} = c - \frac{1}{b}$$

$$\frac{a}{3-a} = \frac{bc-1}{b}$$

$$\frac{3-a}{a} = \frac{b}{bc-1} \quad \text{--- M1}$$

$$\frac{3}{a} - 1 = \frac{b}{bc-1}$$

$$\frac{3}{a} = \frac{b}{bc-1} + 1 \quad \text{--- M1}$$

$$\frac{3}{a} = \frac{b+bc-1}{bc-1}$$

$$\frac{a}{3} = \frac{bc-1}{b+bc-1}$$

$$a = \frac{3(bc-1)}{b+bc-1} \quad \text{--- A1}$$

$$\text{Answer } a = \frac{3bc-3}{b+bc-1} \quad [3]$$





(d) Solve the equation $\frac{x}{2x-1} - \frac{6}{4-x} = 3$.

Give your solutions correct to two decimal places.

$$x(4-x) - 6(2x-1) = 3(2x-1)(4-x) \quad - M1$$

$$4x - x^2 - 12x + 6 = 3(8x - 4 - 2x^2 + x)$$

$$-x^2 - 8x + 6 = -6x^2 + 27x - 12$$

$$5x^2 - 35x + 18 = 0. \quad - M1$$

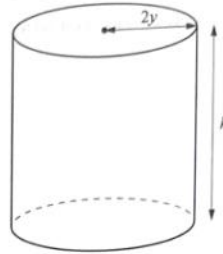
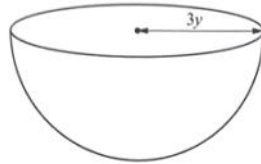
$$x = \frac{-(-35) \pm \sqrt{(-35)^2 - 4(5)(18)}}{2(5)} \quad - M1$$

$$= \frac{35 \pm \sqrt{865}}{10}$$

$$x = 6.44 \text{ or } x = 0.56 \quad \begin{matrix} M1 & M1 \\ | & | \\ 0.56 & 6.44 \end{matrix}$$

(2 d.p.)

Answer $x = 0.56$ or 6.44 [5]



The diagram shows a solid hemisphere and a solid cylinder.
The hemisphere has radius $3y$ cm.
The cylinder has radius $2y$ cm and height h cm.

- (a) Show that the total surface area of the hemisphere is $27\pi y^2 \text{ cm}^2$.

Answer

Total surface area

$$= \pi(3y)^2 + 2\pi(3y)^2 - M1$$

$$= 9\pi y^2 + 18\pi y^2$$

$$= 27\pi y^2 - A1$$

[2]

- (b) The total surface area of the cylinder is equal to the total surface area of the hemisphere.

Find h in terms of y .

$$27\pi y^2 = 2\pi(2y)^2 + 2\pi(2y)(h) - M1$$

$$27\pi y^2 = 8\pi y^2 + 4\pi y h$$

$$19\pi y^2 = 4\pi y h - M1$$

$$h = \frac{19\pi y^2}{4\pi y}$$

$$= \frac{19}{4} y$$

Answer $h = \frac{19}{4} y - A1$ [3]



- (e) The volume of the hemisphere is 500cm^3 .

Calculate the volume of the cylinder.

$$500 = \frac{2}{3} \pi (3y)^3 - M1$$

$$500 = 18\pi y^3.$$

$$y^3 = \frac{250}{9\pi} - M1$$

$$\text{Volume of cylinder} = \pi (2y)^2 (h)$$

$$= 4\pi y^2 \left(\frac{19}{4}y\right) - M1$$

$$= 19\pi y^3$$

$$= 19\pi \times \frac{250}{9\pi} \quad \text{Answer } 527\frac{7}{9} \text{ cm}^3 [4]$$

$$= 527\frac{7}{9} \text{ cm}^3 - A1$$





- 4 (a) Complete the table of values for $y = \frac{2}{x^2} + 3x - 1$.

Values are given to one decimal place where appropriate.

x	-3	-2	-1	-0.5	0.5	1	1.5	2	3
y	-9.8	-6.5	-2	5.5	8.5	4	4.4	5.5	8.2

[1]

- (b) On the grid opposite, draw the graph of $y = \frac{2}{x^2} + 3x - 1$ for $-3 \leq x \leq 3$.

[3]

- (c) (i) On the same grid, draw the graph of $2y - 5x = 7$ for $-3 \leq x \leq 3$.

x	-3	0	3
y	-4	3.5	11

[2]

- (ii) Write down the x -coordinates of the points where the line intersects the curve.

- (iii) These values of x are solutions of the equation $x^3 + Ax^2 + B = 0$.
 Find the value of A and the value of B .

$$y = \frac{2}{x^2} + 3x - 1 \quad (1)$$

$$2y - 5x = 7 \quad (2)$$

Sub (1) into (2):

$$2\left(\frac{2}{x^2} + 3x - 1\right) - 5x = 7 \quad -M1$$

$$\frac{4}{x^2} + 6x - 2 - 5x = 7$$

$$x - 9 + \frac{4}{x^2} = 0$$

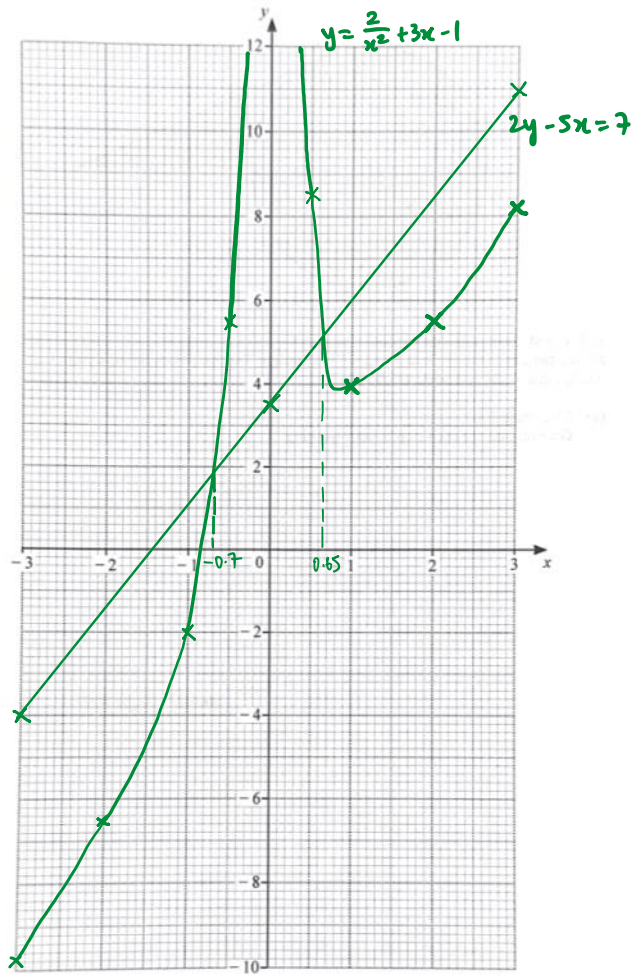
$$x^3 - 9x^2 + 4 = 0$$

Answer $A = -9$ $-A1$
 $B = 4$ $-A1$

[3]



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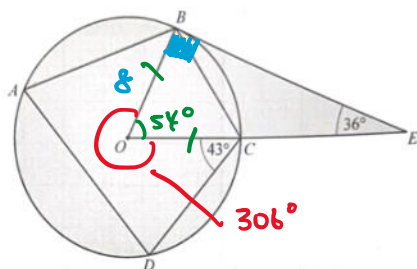




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A, B, C and D are points on the circle, centre O .
 EB is a tangent to the circle at B and OE is a straight line that passes through point C .
Angle $OEB = 36^\circ$ and angle $OCD = 43^\circ$.

- (a) Find angle BAD .
Give reasons for each step of your working.

$$\begin{aligned}\angle OBE &= 90^\circ \text{ (tangent perpendicular to radius)} \\ \angle BOE &= 180^\circ - 90^\circ - 36^\circ \text{ (angle sum of } \triangle BOE) \\ &= 54^\circ\end{aligned}$$

$$\begin{aligned}\angle BCO &= \frac{180^\circ - 54^\circ}{2} \text{ (base angles of isos. } \triangle) \\ &= 63^\circ\end{aligned}$$

Answer Angle $BAD = 74^\circ$ [4]

$$\begin{aligned}\angle BAD &= 180^\circ - 63^\circ - 43^\circ \text{ (angles in opp. segment)} \\ &= 74^\circ\end{aligned}$$





- (b) The radius of the circle is 8 cm.

Calculate the perimeter of the shaded shape.

$$\frac{8}{BE} = \tan 36^\circ$$

$$BE = \frac{8}{\tan 36^\circ}$$

$$\frac{8}{OE} = \sin 36^\circ$$

$$OE = \frac{8}{\sin 36^\circ}$$

$$CE = \frac{8}{\sin 36^\circ} - 8$$

Perimeter

$$= \frac{306}{360} \times 2\pi(8) + \frac{8}{\tan 36^\circ} + \frac{8}{\sin 36^\circ} - 8$$

$$= 59.3 \text{ cm (3 s.f.)}$$

Answer 59.3 cm [4]





- 6 (a) $\mathbf{p} = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$.

X is the point $(7, 3)$ and Y is the point $(3, 4)$.

$\vec{XY} = m\mathbf{p} + n\mathbf{q}$, where m and n are constants.

Find the value of m and the value of n .

$$\begin{aligned}\vec{XY} &= \vec{XO} + \vec{OY} \\ &= -\vec{OX} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} \\ &= -\begin{pmatrix} 7 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} -4 \\ 1 \end{pmatrix} \quad \text{--- M1}\end{aligned}$$

$$\begin{pmatrix} -4 \\ 1 \end{pmatrix} = m \begin{pmatrix} -4 \\ 5 \end{pmatrix} + n \begin{pmatrix} 2 \\ -3 \end{pmatrix} \quad \text{--- M1}$$

$$\begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -4m + 2n \\ 5m - 3n \end{pmatrix}$$

$$-4m + 2n = -4$$

$$n = -2 + 2m - (1)$$

$$1 = 5m - 3n - (2)$$

Sub (1) into (2):

$$1 = 5m - 3(-2 + 2m) \quad \text{--- M1}$$

$$1 = 5m + 6 - 6m$$

$$m = 5$$

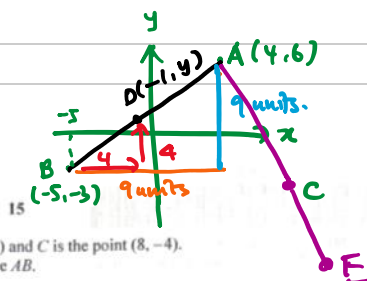
$$n = 8$$

$$\text{Answer } m = \underline{5}$$

$$n = \underline{8}$$

3A1

[4]



- (b) A is the point $(4, 6)$, B is the point $(-5, -3)$ and C is the point $(8, -4)$.
 D is the point $(-1, y)$ which lies on the line AB .

- (i) Find the value of y .

$$\begin{aligned}y &= -3 + 4 \\ &= 1.\end{aligned}$$

$$\text{Answer } y = \underline{1} \quad [1]$$

- (ii) Find the length of the line DC .

$$\text{length} = \sqrt{(8+1)^2 + (-4-1)^2} \quad \text{--- M1}$$

$$= 10.3 \text{ units} \quad (3 \text{ s.f.})$$

$$10.3 \text{ units}$$

$$\text{Answer } \underline{\hspace{2cm}} \quad [2]$$

$$= 10.3 \text{ units} \\ (3 \text{ s.f.})$$

$$10.3 \text{ units} \quad \text{A1}$$

Answer [2]

(iii) C is the point on the line AE such that AC : AE = 2 : 3.

Find the equation of the line BE.

$$\begin{aligned} \vec{AC} &= \vec{OC} - \vec{OA} \\ &= \begin{pmatrix} 8 \\ -4 \end{pmatrix} - \begin{pmatrix} 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ -10 \end{pmatrix} \end{aligned}$$

$$\vec{AE} = \frac{3}{2} \begin{pmatrix} 4 \\ -10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix} \quad \text{--- M1}$$

$$\vec{OE} = \vec{OA} + \vec{AE} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \begin{pmatrix} 6 \\ -15 \end{pmatrix} = \begin{pmatrix} 10 \\ -9 \end{pmatrix} \quad \text{--- M1}$$

$$\begin{aligned} m_{BE} &= \frac{-9 - (-3)}{10 - (-5)} \\ &= -\frac{2}{5} \quad \text{--- M1} \end{aligned}$$

$$\text{eqn: } \frac{y+3}{x+5} = -\frac{2}{5}$$

$$5y+15 = -2x-10$$

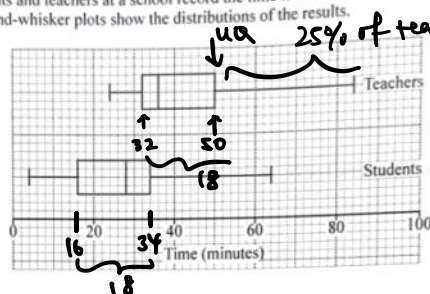
$$5y+2x+25=0$$

$$\text{Answer } 5y+2x+25=0 \quad \text{A1.} \quad [4]$$

(accept: $y = -\frac{2}{5}x - 5$)



- 7 (a) The students and teachers at a school record the time it took them to travel to school. The box-and-whisker plots show the distributions of the results.



- (i) There are 18 teachers who took more than 50 minutes to travel to school.

Work out the number of teachers in the school.

$$18 \times \frac{100}{25} = 72 \quad \text{Answer } \dots\dots\dots 72 \text{ --- BI} \quad [1]$$

- (ii) Make a comment comparing the averages and a comment comparing the distributions of the times taken by the students and the teachers. Use figures to support your answers.

- BI: comment
BI: 18 min for IQR
1. On average, teachers took a longer time to travel as median travelling time for teachers is 36 minutes, which is greater than median time for children at 28 minutes.
2. The time taken to travel to school for students and teachers is equally spread out as the interquartile range for both groups is at 18 minutes. [3]

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(b) In a class of 24 students, 8 walk to school, 6 cycle to school and the rest travel to school by bus.

(i) One student is selected from the class at random.

Find the probability that the student travels to school by bus.

$$\frac{10}{24} = \frac{5}{12} \quad \text{Answer } \frac{5}{12} \text{ - BI [1]}$$

(ii) Two students are selected from the class at random.

Find the probability that they both use the same type of transport to travel to school.

$$P(\text{same}) = \frac{8}{24} \times \frac{7}{23} + \frac{6}{24} \times \frac{5}{23} + \frac{10}{24} \times \frac{9}{23} \text{ - MI: any 2 correct.}$$
$$= \frac{22}{69}$$

$$\text{Answer } \frac{22}{69} \text{ - AI [2]}$$

(iii) Three of the students are selected from the class at random.

Find the probability that two of them walk to school and the other does not.

$$\text{Probability} = P(WWDW) + P(DW WW) + P(WDW W)$$
$$= \frac{8}{24} \times \frac{7}{23} \times \frac{16}{22} \times 3 \text{ - MI}$$
$$= \frac{56}{253} \quad \text{Answer } \frac{56}{253} \text{ - AI [2]}$$



- 8 (a) The diagram shows part of a number grid.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40

A cross outlining five numbers, as shown, is placed on the grid.

- (i) The cross is placed in a different position on the grid.
The sum of the largest and smallest of the numbers outlined is 204.

Complete the cross to show the numbers outlined.

Let smallest no. be x .

$$x + (x + 16) = 204.$$

$$2x = 188$$

$$x = 94.$$

Answer

	94	
101	102	103
	110	

[3]

- (ii) The cross can be placed anywhere on the grid.
The difference between the product of the left and right numbers and the product of the top and bottom numbers is found.

Show that this difference is always 63.

Answer

Let x be top number.

left $\times$ right - top $\times$ bottom

$$= (x+7)(x+9) - x(x+16)$$

$$= x^2 + 16x + 63 - x^2 - 16x$$

$$= 63 \text{ (shown)} - A1$$





- (b) The n th term of a sequence is given by $T_n = \frac{n^2 - 1}{3n - 1}$.

The term $T_k = \frac{15}{4}$.

Find the value of T_{k+1} .

$$\frac{15}{4} = \frac{k^2 - 1}{3k - 1}$$

$$45k - 15 = 4k^2 - 4$$

$$4k^2 - 45k + 11 = 0$$

$$(4k - 1)(k - 11) = 0$$

$$k = \frac{1}{4} \text{ or } k = 11.$$

(reject)

$$T_{k+1} = T_{12}$$

$$= \frac{12^2 - 1}{3(12) - 1}$$

$$= 4\frac{3}{35}$$

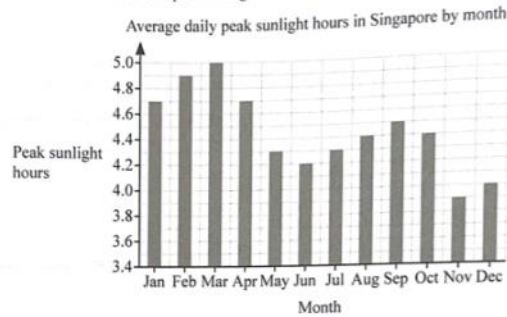
Answer $T_{k+1} = 4\frac{3}{35}$ [4]



- 9 Chen is considering installing solar panels at his home. He wants to estimate the area of solar panels he will need to generate the amount of electricity used by his household. He finds this information on the internet.

- Find the average amount of electricity, measured in kilowatt hours (kWh), used in one day
- Find the number of peak sunlight hours you receive per day
- Solar panel output in daily watt hours = solar panel watts $\times$ peak sunlight hours $\times$ 75%
- All solar panels should be of the same type
- 1 kWh is the amount of energy used in 1 hour by an appliance with a power rating of 1 kW
- 1 kW = 1000 W

He does some more research about peak sunlight hours and the specifications of common solar panels.



Type	Solar panel wattage	Length of panel	Width of panel	Mass of panel
A	300 W	1.46 m	1.05 m	18 kg
B	370 W	1.96 m	1.00 m	23 kg
C	390 W	1.98 m	1.00 m	28 kg
D	400 W	1.69 m	1.05 m	19 kg

- (a) A games console has a power rating of 165 W.

Calculate the amount of energy, in kWh, used in 5 hours by the games console.

$$\frac{165}{1000} \times 5 = 0.825 \text{ kWh}$$

Answer 0.825 kWh [1]

- (b) Assuming all months have an equal number of days, calculate the average daily peak sunlight hours in Singapore over the 12-month period.

$$\text{avg. daily peak} = 4.4416$$

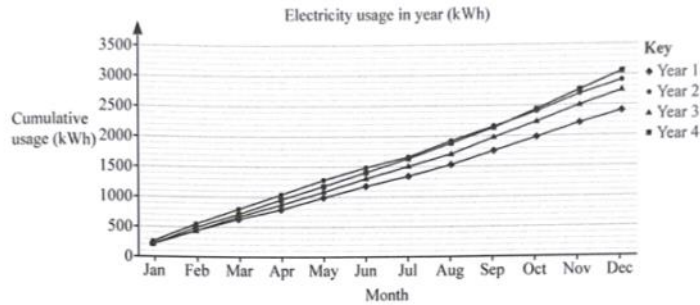
$$= 4.44 \text{ hours (3s.f.)}$$

Answer 4.44 hours [2]





(c) This graph shows the electricity used by Chen's household for the past four years.



Find an estimate of the total area of solar panels Chen will need for his home.
Justify any decisions you make and show your calculations clearly.

M1 } To ensure Chen has sufficient energy, use
highest cumulative usage (i.e. Year 4),
which is 3050 kWh

$$\text{Avg. daily energy usage} = \frac{3050}{365} = 8.3561 \text{ kWh} \quad \text{M1}$$
$$= 8356.1 \text{ Wh.}$$

Type	Solar panel output	No. of solar panels	Area (m ²)
A	$300 \times 4.4416 \times \frac{75}{100} = 999.36$	$8356.1 \div 999.36 \approx 9$	13.797
B	$370 \times 4.4416 \times 0.75 = 1232.544$	$8356.1 \div 1232.544 \approx 7$	13.72
C	$390 \times 4.4416 \times 0.75 = 1299.168$	$8356.1 \div 1299.168 \approx 7$	13.86
D	$400 \times 4.4416 \times 0.75 = 1332.48$	$8356.1 \div 1332.48 \approx 7$	12.4215

Choose Type D for smallest total
area of solar panels at 12.5 m² (3 s.f.)
round up

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